# Supplementary Material S1

A Critical Review of Evaluation Evidence for Autonomous Driving World Models

Release 1.0.0 | Prepared 2 November 2026 | 17 systems | 35 task/input records | 4 numerical contrasts

## Scope and provenance

This newly assembled supplement reconstructs the register described in Revision 4 using the supplied manuscript and targeted primary-source checking. The original S1, historical extraction log and original script were not supplied. This AI-assisted reconstruction does not establish that the same files or workflow existed earlier.

The sample is purposive. Task records are not independent studies. No independent second-coder audit, model-training replication, new driving rollout or physical-road validation is claimed. Missing quality fields mean not extracted, not necessarily absent from a paper.

## Retrieval and extraction

The 17 named seed systems were retrieved through exact-title primary records, official proceedings and linked author manuscripts. Relevant methods, action interfaces and endpoint-specific evaluation passages were checked. Inspected versions control the locators below; equivalence to a later publication is not assumed. Present reconstruction operations are recorded in data/retrieval_provenance.json.

The unit is study x task/input variant x specified claim. Source-reported endpoints are paraphrased. Stronger use questions introduced by this review are recorded separately and must not be attributed to the original authors.

## Codebook

| Axis | Meaning |
|---|---|
| R | Predicted representation or decoded observation |
| A | External action conditions and ego outputs; record future information separately |
| I | Mechanism producing other-agent response, not proof of response validity |
| U | Stage at which prediction is used |
| V | Reference and feedback process actually tested |

| Status | Claim-specific meaning |
|---|---|
| D | Direct test of the specified endpoint; not a quality score |
| P | Proxy, selected qualitative evidence, or indirect association for the specified claim |
| X | The inspected test addresses a different endpoint; not proof of incapability |
| NR | Unresolved in the inspected material; not a negative result |

D is directness, not high experimental quality. X is not incapability and NR is not a negative result. No total maturity score is assigned.

# Source index

| ID | System | Inspected version |
|---|---|---|
| S01 | DriveDreamer | arXiv:2309.09777v2 |
| S02 | Drive-WM | arXiv:2311.17918v1 |
| S03 | Vista | arXiv:2405.17398v5 |
| S04 | GenAD | arXiv:2403.09630v2 |
| S05 | OccWorld | ECCV 2024 open-access proceedings PDF |
| S06 | Copilot4D | arXiv:2311.01017v4 |
| S07 | MILE | arXiv:2210.07729v2 |
| S08 | Think2Drive | arXiv:2402.16720v2 |
| S09 | UniSim | arXiv:2308.01898v1 |
| S10 | NeuRAD | arXiv:2311.15260v3 |
| S11 | DriveArena | arXiv:2408.00415v1 |
| S12 | LAW | ICLR 2025 proceedings PDF including appendix |
| S13 | World4Drive | arXiv:2507.00603v1 |
| S14 | WoTE | arXiv:2504.01941v2 |
| S15 | WorldRFT | arXiv:2512.19133v1 |
| S16 | TrafficBots | arXiv:2303.04116v2 |
| S17 | SceneDiffuser | arXiv:2412.12129v1 |

# Task and input register

## S01 DriveDreamer

DriveDreamer: Towards Real-world-driven World Models for Autonomous Driving. Manuscript reference [50].

Primary source: arXiv:2309.09777v2

### S01-T01 Conditional video generation

| Field | Recorded interface |
|---|---|
| R | video |
| A | structured scene and driving-action conditions |
| I | generated observations |
| U | generation |
| V | logged; nuScenes |

Locator: Sections 3, 4.2.

Source endpoint (paraphrased): Conditional video quality. Status D.

Additional review question: Independent physical action-outcome ordering. Status NR.

Boundary: Image agreement does not label executed alternatives.

**S01-T02 Action prediction head**

| Field | Recorded interface |
|---|---|
| R | latent features and ego actions |
| A | past actions; future actions output |
| I | no responsive execution in this test |
| U | deployment |
| V | logged; nuScenes |

Locator: Section 4.3; Figure 3.

Source endpoint (paraphrased): Logged planning accuracy. Status D.

Additional review question: Reactive policy execution from this logged test. Status X.

Boundary: Planning head combines action history and visual features.

## S02 Drive-WM

Driving into the Future: Multiview Visual Forecasting and Planning with World Model for Autonomous Driving. Manuscript reference [49].

Primary source: arXiv:2311.17918v1

**S02-T01 Tree-based candidate planning**

| Field | Recorded interface |
|---|---|
| R | multiview video |
| A | supplied ego trajectory candidates |
| I | generated candidate futures |
| U | online_selection |
| V | logged; nuScenes |

Locator: Sections 4.1-4.2, 5.4.

Source endpoint (paraphrased): Logged candidate-selection planning. Status D.

Additional review question: Independently realized candidate-outcome ranking. Status P.

Boundary: Image-based reward supports selection; independent cost labels remain distinct.

### S02-T02 Counterfactual and deviation examples

| Field | Recorded interface |
|---|---|
| R | multiview video |
| A | edited ego trajectories and scene conditions |
| I | generated alternative futures |
| U | generation |
| V | qualitative; edited recorded scenes |

Locator: Sections 5.4-5.5; Appendix A.4.

Source endpoint (paraphrased): Selected counterfactual examples. Status P.

Additional review question: Physical response distribution calibration. Status P.

Boundary: Selected examples are proxy evidence for physical response.

## S03 Vista

Vista: A Generalizable Driving World Model with High Fidelity and Versatile Controllability. Manuscript reference [51].

Primary source: arXiv:2405.17398v5

### S03-T01 Action-conditioned future generation

| Field | Recorded interface |
|---|---|
| R | video |
| A | command, goal, trajectory, steering or speed |
| I | generated observations |
| U | generation |
| V | logged; recorded driving datasets |

Locator: Sections 3.2, 4.1-4.2.

Source endpoint (paraphrased): Forecast and control-adherence diagnostics. Status D.

Additional review question: Response calibration under ego intervention. Status NR.

Boundary: Different control modalities define different conditional tasks.

#### S03-T02 Reward on perturbed trajectories

| Field | Recorded interface |
|---|---|
| R | video-derived reward |
| A | jittered recorded trajectories |
| I | model-generated futures |
| U | action_scoring |
| V | logged; Waymo; perturbation experiment |

Locator: Sections 3.3, 4.3; Appendix C.6.

Source endpoint (paraphrased): Reward association with trajectory perturbations. Status D.

Additional review question: Independent hazardous-outcome ranking. Status P.

Boundary: Perturbations do not independently execute every alternative.

### S04 GenAD

Generalized Predictive Model for Autonomous Driving. Manuscript reference [33].

Primary source: arXiv:2403.09630v2

#### S04-T01 Trajectory-conditioned forecast adaptation

| Field | Recorded interface |
|---|---|
| R | video |
| A | supplied future trajectory |
| I | generated observations |
| U | generation |
| V | logged; nuScenes |

Locator: Section 3.3; Appendix D.1.2.

Source endpoint (paraphrased): Conditional future prediction. Status D.

Additional review question: Independent reaction-probability calibration. Status NR.

Boundary: The trajectory is a condition.

#### S04-T02 Frozen-encoder motion planner

| Field | Recorded interface |
|---|---|
| R | latent scene features |
| A | history and command; future waypoints output |
| I | planning head without candidate rollout search |
| U | representation_transfer |
| V | logged; nuScenes |

Locator: Section 3.3; Table 5; Appendix D.1.3.

Source endpoint (paraphrased): Frozen-encoder planning transfer. Status D.

Additional review question: Inference-time candidate search in this adaptation. Status X.

Boundary: A frozen encoder and MLP implement the planning adaptation.

### S05 OccWorld

OccWorld: Learning a 3D Occupancy World Model for Autonomous Driving. Manuscript reference [48].

Primary source: ECCV 2024 open-access proceedings PDF

#### S05-T01 Oracle occupancy

| Field | Recorded interface |
|---|---|
| R | 3D occupancy and ego motion |
| A | ego trajectory jointly output |
| I | joint predicted scene and ego evolution |
| U | forecast_and_planning |
| V | logged; nuScenes |

Locator: Tables 1-2.

Information: Ground-truth 3D occupancy input.

Source endpoint (paraphrased): Logged occupancy forecasting and planning. Status D.

Additional review question: External ego-intervention validity. Status NR.

Boundary: Joint ego output is not an externally supplied intervention.

**S05-T02 Camera OccWorld-D**

| Field | Recorded interface |
|---|---|
| R | 3D occupancy and ego motion |
| A | ego trajectory jointly output |
| I | joint predicted scene and ego evolution |
| U | forecast_and_planning |
| V | logged; nuScenes |

Locator: Tables 1-2.

Information: Camera input with 3D-occupancy supervision.

Source endpoint (paraphrased): Logged occupancy forecasting and planning. Status D.

Additional review question: External ego-intervention validity. Status NR.

Boundary: Joint ego output is not an externally supplied intervention.

**S05-T03 Camera OccWorld-T**

| Field | Recorded interface |
|---|---|
| R | 3D occupancy and ego motion |
| A | ego trajectory jointly output |
| I | joint predicted scene and ego evolution |
| U | forecast_and_planning |
| V | logged; nuScenes |

Locator: Tables 1-2.

Information: Camera input with semantic-LiDAR supervision.

Source endpoint (paraphrased): Logged occupancy forecasting and planning. Status D.

Additional review question: External ego-intervention validity. Status NR.

Boundary: Joint ego output is not an externally supplied intervention.

**S05-T04 Camera OccWorld-S**

| Field | Recorded interface |
|---|---|
| R | 3D occupancy and ego motion |
| A | ego trajectory jointly output |
| I | joint predicted scene and ego evolution |
| U | forecast_and_planning |
| V | logged; nuScenes |

Locator: Tables 1-2.

Information: Camera input without additional scene supervision.

Source endpoint (paraphrased): Logged occupancy forecasting and planning. Status D.

Additional review question: External ego-intervention validity. Status NR.

Boundary: Joint ego output is not an externally supplied intervention.

**S05-T05 Tokenizer resolution ablation**

| Field | Recorded interface |
|---|---|
| R | 3D occupancy |
| A | joint ego output; tokenizer varied |
| I | joint predicted scene and ego evolution |
| U | ablation |
| V | logged; nuScenes; OccWorld-O |

Locator: Table 3, PDF page 13.

Information: Oracle occupancy; 50x50 versus 100x100 tokenizer.

Source endpoint (paraphrased): Reconstruction, forecasting and planning contrast. Status D.

Additional review question: Universal reconstruction-to-driving conversion. Status X.

Boundary: Tokenizer resolution and computational burden change.

## S06 Copilot4D

Copilot4D: Learning Unsupervised World Models for Autonomous Driving via Discrete Diffusion. Manuscript reference [30].

Primary source: arXiv:2311.01017v4

**S06-T01 Future-pose-conditioned point-cloud forecast**

| Field | Recorded interface |
|---|---|
| R | point clouds |
| A | supplied future ego poses |
| I | learned conditional scene evolution |
| U | forecast |
| V | logged; nuScenes, KITTI Odometry, Argoverse2 |

Locator: Section 5; Appendix A.2.2.

Information: Past point clouds and future ego poses.

Source endpoint (paraphrased): Conditional point-cloud prediction. Status D.

Additional review question: Unconditioned ego-action prediction. Status X.

Boundary: Future ego poses are supplied evaluation inputs.

#### S06-T02 Edited-action response visualization

| Field | Recorded interface |
|---|---|
| R | point clouds |
| A | edited future ego trajectories |
| I | generated surrounding-agent motion |
| U | generation |
| V | qualitative; selected scenes |

Locator: Appendix A.7; Figure 12.

Source endpoint (paraphrased): Counterfactual visualization. Status P.

Additional review question: Physical response validity. Status P.

Boundary: A selected braking example is not a calibrated response distribution.

## S07 MILE

Model-Based Imitation Learning for Urban Driving. Manuscript reference [52].

Primary source: arXiv:2210.07729v2

#### S07-T01 Camera-based policy execution

| Field | Recorded interface |
|---|---|
| R | latent state; decoded BEV |
| A | camera history; policy actions output |
| I | responsive CARLA traffic outside learned transitions |
| U | policy_learning |
| V | reactive_simulation; CARLA held-out town and weather |

Locator: Sections 4, 5.1, 5.3, 6.2.

Source endpoint (paraphrased): Reactive policy performance. Status D.

Additional review question: Physical-road transfer. Status NR.

Boundary: Simulation execution is positive evidence conditional on CARLA.

## S08 Think2Drive

Think2Drive: Efficient Reinforcement Learning by Thinking with Latent World Model for Autonomous Driving (in CARLA-v2). Manuscript reference [53].

Primary source: arXiv:2402.16720v2

### S08-T01 Privileged-input policy execution

| Field | Recorded interface |
|---|---|
| R | latent state and BEV |
| A | privileged scene state; throttle, steer, brake |
| I | responsive CARLA traffic outside learned transitions |
| U | policy_learning |
| V | reactive_simulation; CARLA Leaderboard v2 |

Locator: Sections 3.1-3.2, 4.4; Appendix 0.B.

Information: Privileged boxes, map and traffic signals.

Source endpoint (paraphrased): Reactive planning performance. Status D.

Additional review question: Raw-camera deployment from this privileged-input test. Status X.

Boundary: Planning is isolated using simulator scene information.

## S09 UniSim

UniSim: A Neural Closed-Loop Sensor Simulator. Manuscript reference [39].

Primary source: arXiv:2308.01898v1

### S09-T01 Edited-pose sensor simulation

| Field | Recorded interface |
|---|---|
| R | camera and lidar |
| A | sensor poses and actor placements |
| I | sensor rendering; behavior external |
| U | sensor_simulation |
| V | sensor_reference; recorded driving scenes |

Locator: Sections 3, 4.3-4.4.

Source endpoint (paraphrased): Sensor realism and downstream domain-gap diagnostics. Status D.

Additional review question: Learned traffic law established by renderer. Status X.

Boundary: Renderer and behavior mechanism are separate components.

#### S09-T02 Full-autonomy simulation demonstration

| Field | Recorded interface |
|---|---|
| R | camera and lidar |
| A | autonomy actions and edited actors |
| I | heuristic or human-specified actor trajectories |
| U | hybrid_simulation |
| V | hybrid_execution; UniSim full-autonomy setup |

Locator: Section 4.5; Appendix A1 actor behavior.

Source endpoint (paraphrased): Full-autonomy simulation demonstrations. Status D.

Additional review question: Physical response probability calibration. Status NR.

Boundary: Counterfactual actor motion is external to neural sensor rendering.

## S10 NeuRAD

NeuRAD: Neural Rendering for Autonomous Driving. Manuscript reference [40].

Primary source: arXiv:2311.15260v3

#### S10-T01 Novel-view and novel-scenario rendering

| Field | Recorded interface |
|---|---|
| R | camera and lidar |
| A | sensor and actor poses |
| I | rendering of supplied scene trajectories |
| U | sensor_simulation |
| V | sensor_reference; five driving datasets |

Locator: Sections 3, 4.2-4.3; Appendices F-G.

Source endpoint (paraphrased): Novel-view sensor quality. Status D.

Additional review question: Traffic-response validity from rendering tests. Status X.

Boundary: Pose and sensor modeling do not constitute a learned traffic policy.

## S11 DriveArena

DriveArena: A Closed-loop Generative Simulation Platform for Autonomous Driving. Manuscript reference [41].

Primary source: arXiv:2408.00415v1

### S11-T01 Generative-image feedback loop

| Field | Recorded interface |
|---|---|
| R | camera images |
| A | agent trajectories; scene layouts |
| I | external Traffic Manager with World Dreamer |
| U | hybrid_simulation |
| V | hybrid_execution; DriveArena |

Locator: Sections 3.1-3.4, 4.5.

Source endpoint (paraphrased): Open-loop and closed-loop agent evaluation. Status D.

Additional review question: Physical ordering of candidate policies. Status NR.

Boundary: Behavior originates in the external traffic component.

## S12 LAW

Enhancing End-to-End Autonomous Driving with Latent World Model. Manuscript reference [34].

Primary source: ICLR 2025 proceedings PDF including appendix

### S12-T01 Perception-free latent supervision

| Field | Recorded interface |
|---|---|
| R | future visual latents |
| A | current latent and predicted ego trajectory |
| I | auxiliary future-feature prediction |
| U | training_auxiliary |
| V | logged; nuScenes |

Locator: Table 4, PDF page 9.

Information: Perception-free framework.

Source endpoint (paraphrased): Planning benefit of latent supervision. Status D.

Additional review question: Inference-time search attribution. Status X.

Boundary: Visual latents and trajectories are inputs, not two predicted outputs.

**S12-T02 Perception-based latent supervision**

| Field | Recorded interface |
|---|---|
| R | future visual latents |
| A | current latent and predicted ego trajectory |
| I | auxiliary future-feature prediction |
| U | training_auxiliary |
| V | logged; nuScenes |

Locator: Table 4, PDF page 9.

Information: Perception-based framework.

Source endpoint (paraphrased): Planning benefit with perception framework. Status D.

Additional review question: Search attribution from auxiliary ablation. Status X.

Boundary: Additional perception supervision changes the input and training setting.

**S12-T03 NAVSIM auxiliary-prediction ablation**

| Field | Recorded interface |
|---|---|
| R | future visual latents |
| A | action-conditioned auxiliary task |
| I | non-reactive environment scoring |
| U | training_auxiliary |
| V | nonreactive_simulation; NAVSIM |

Locator: Table 5, PDF page 9.

Source endpoint (paraphrased): PDMS with and without latent prediction. Status D.

Additional review question: Reactive traffic utility from this NAVSIM test. Status X.

Boundary: This is a training ablation, not a runtime candidate-search test.

**S12-T04 CARLA auxiliary-prediction ablation**

| Field | Recorded interface |
|---|---|
| R | future visual latents |
| A | policy trajectories output |
| I | responsive CARLA traffic |
| U | training_auxiliary |
| V | reactive_simulation; CARLA Town05 Long |

Locator: Tables 3, 5.

Source endpoint (paraphrased): Driving-score ablation. Status D.

Additional review question: Physical-road transfer. Status NR.

Boundary: Reactive utility is separate from inference-time search.

## S13 World4Drive

World4Drive: End-to-End Autonomous Driving via Intention-aware Physical Latent World Model. Manuscript reference [35].

Primary source: arXiv:2507.00603v1

### S13-T01 Intention-aware selection on logs

| Field | Recorded interface |
|---|---|
| R | future world latents |
| A | intention-conditioned trajectory candidates |
| I | future latent prediction and learned selector |
| U | online_selection |
| V | logged; nuScenes |

Locator: Sections 3.3.2, 4.3.

Source endpoint (paraphrased): Intention-aware planning. Status D.

Additional review question: Hazardous-outcome cost calibration. Status P.

Boundary: Training labels use actual-future latent distance; inference uses ScoreNet.

### S13-T02 Intention-aware selection in NAVSIM

| Field | Recorded interface |
|---|---|
| R | future world latents |
| A | intention-conditioned trajectory candidates |
| I | future latent prediction and learned selector |
| U | online_selection |
| V | nonreactive_simulation; NAVSIM |

Locator: Sections 3.3.2, 4.1, 4.3.

Source endpoint (paraphrased): Non-reactive planning score. Status D.

Additional review question: Reactive traffic execution from this NAVSIM test. Status X.

Boundary: The source closed-loop label does not imply responsive other-agent execution.

## S14 WoTE

End-to-End Driving with Online Trajectory Evaluation via BEV World Model. Manuscript reference [36].

Primary source: arXiv:2504.01941v2

### S14-T01 Future-state trajectory evaluator in NAVSIM

| Field | Recorded interface |
|---|---|
| R | future BEV features |
| A | trajectory candidates |
| I | learned future states with reward evaluation |
| U | online_selection |
| V | nonreactive_simulation; NAVSIM |

Locator: Section 4.4; Table 3.

Source endpoint (paraphrased): Future-state evaluator ablation. Status D.

Additional review question: Road benefit from this non-reactive ablation. Status X.

Boundary: Fix evaluator presence when contrasting future-state inputs.

### S14-T02 Future-state evaluation in CARLA integration

| Field | Recorded interface |
|---|---|
| R | future BEV features |
| A | trajectory candidates |
| I | responsive CARLA execution |
| U | online_selection |
| V | reactive_simulation; CARLA Bench2Drive |

Locator: Sections 4.2-4.3; Table 2.

Source endpoint (paraphrased): Integrated reactive driving utility. Status D.

Additional review question: Physical-road transfer. Status NR.

Boundary: Integration supports utility; matched-resource attribution remains separate.

## S15 WorldRFT

WorldRFT: Latent World Model Planning with Reinforcement Fine-Tuning for Autonomous Driving. Manuscript reference [54].

Primary source: arXiv:2512.19133v1

### S15-T01 Supervised versus reinforcement fine-tuning

| Field | Recorded interface |
|---|---|
| R | world latents and planned trajectories |
| A | planning outputs; reinforcement refinement |
| I | policy optimization under logged evaluation |
| U | policy_refinement |
| V | logged; nuScenes |

Locator: Additional Experimental Results; Table 11.

Source endpoint (paraphrased): SFT/RFT L2 and collision endpoints. Status D.

Additional review question: Transition-accuracy improvement isolated by this ablation. Status X.

Boundary: Equal rounded L2 does not imply equal unrounded errors.

### S15-T02 NAVSIM planning evaluation

| Field | Recorded interface |
|---|---|
| R | world latents and planned trajectories |
| A | planning outputs |
| I | non-reactive environment scoring |
| U | policy_refinement |
| V | nonreactive_simulation; NAVSIM |

Locator: Experiment; Benchmark and Main Results.

Source endpoint (paraphrased): Non-reactive planning score. Status D.

Additional review question: Reactive traffic claim from this NAVSIM test. Status X.

Boundary: NAVSIM is not evidence of responding traffic.

## S16 TrafficBots

TrafficBots: Towards World Models for Autonomous Driving Simulation and Motion Prediction. Manuscript reference [37].

Primary source: arXiv:2303.04116v2

### S16-T01 Motion-prediction evaluation

| Field | Recorded interface |
|---|---|
| R | agent states |
| A | destinations and latent personalities |
| I | multi-agent policies |
| U | traffic_simulation |
| V | logged; Waymo Open Motion Dataset |

Locator: Sections III-V.

Source endpoint (paraphrased): Motion-prediction diagnostics. Status D.

Additional review question: Physical intervention calibration. Status NR.

Boundary: Prediction surrogate does not label unseen physical alternatives.

### S16-T02 Multi-agent simulation diagnostics

| Field | Recorded interface |
|---|---|
| R | agent states |
| A | player and bot states; bot conditions |
| I | learned reactive bot policies |
| U | traffic_simulation |
| V | behavior_rollout; Waymo-based rollouts |

Locator: Sections III-V; Appendix F.

Information: Destination/personality conditioning; consult source for rollout mode.

Source endpoint (paraphrased): Behavior and rollout diagnostics. Status D.

Additional review question: Physical response distribution calibration. Status NR.

Boundary: A priori and posterior settings must retain their future-information conditions.

## S17 SceneDiffuser

SceneDiffuser: Efficient and Controllable Driving Simulation Initialization and Rollout. Manuscript reference [38].

Primary source: arXiv:2412.12129v1

### S17-T01 Constrained scene initialization

| Field | Recorded interface |
|---|---|
| R | agent layouts |
| A | global context and hard constraints |
| I | scene-level diffusion |
| U | scene_initialization |
| V | logged; Waymo-based scene generation |

Locator: Sections 3.3-3.4, 4.2.

Source endpoint (paraphrased): Scene-generation and constraint diagnostics. Status D.

Additional review question: Ego-intervention response inferred from initialization. Status X.

Boundary: Initialization constraints do not by themselves test policy response.

### S17-T02 Amortized scene rollout

| Field | Recorded interface |
|---|---|
| R | agent states |
| A | history, context and logged validity mask |
| I | unified AV and agent rollout |
| U | traffic_simulation |
| V | behavior_rollout; WOSAC-style evaluation |

Locator: Sections 3.2, 4.1; Appendix A.3; Limitations.

Information: Logged future validity mask is supplied.

Source endpoint (paraphrased): Traffic-rollout diagnostics. Status D.

Additional review question: Independent ego-intervention response calibration. Status NR.

Boundary: Logged validity and unified AV/agent update qualify intervention inference.

# Numerical transfer register

These are author-reported within-study contrasts, not new experiments. Percent metrics use a 0-100 scale. Absolute changes in these metrics are percentage-point changes. Displayed rounding is retained; no missing interval or significance result is inferred.

## C01 OccWorld

ECCV 2024 proceedings PDF | Table 3, PDF page 13

Dataset and inputs: nuScenes; oracle 3D occupancy; OccWorld-O.

Baseline: (50^2, 128, 512) tokenizer.

Comparison: (100^2, 128, 512) tokenizer.

| Metric | Unit | Baseline | Comparison |
|---|---|---|---|
| reconstruction_mIoU | percent | 66.38 | 78.12 |
| forecast_mIoU | percent | 17.14 | 12.38 |
| planning_L2 | m | 1.17 | 1.36 |

Aggregation: Forecast mIoU and L2 averaged over 1s, 2s, 3s.

Boundary: Tokenizer resolution, token count and compute change together; no universal causal claim.

## C02 LAW

ICLR 2025 proceedings PDF | Table 4, PDF page 9

Dataset and inputs: nuScenes.

Baseline: Perception-free; no latent prediction.

Comparison: Perception-free; visual latent and predicted trajectory supplied to latent predictor.

| Metric | Unit | Baseline | Comparison |
|---|---|---|---|
| planning_L2 | m | 0.71 | 0.61 |
| logged_collision | percent | 0.41 | 0.3 |

Aggregation: Avg. columns over 1s, 2s, 3s.

Boundary: Visual latents and predicted trajectory are model inputs. This is auxiliary supervision, not runtime search.

## C03 WoTE

ICCV 2025 proceedings; cross-checked against arXiv:2504.01941 | Table 3

Dataset and inputs: NAVSIM.

Baseline: Trajectory evaluator present; future states absent.

Comparison: Trajectory evaluator present; future states present.

| Metric | Unit | Baseline | Comparison |
|---|---|---|---|
| PDMS | score_points | 83.2 | 85.6 |

Aggregation: Author-reported PDMS; no reconstructed sampling interval.

Boundary: Non-reactive ablation; does not isolate road effects or every training/compute change.

### C04 WorldRFT

arXiv:2512.19133v1 including supplementary tables | Additional Experimental Results, Table 11

Dataset and inputs: nuScenes.

Baseline: Supervised fine-tuning (SFT).

Comparison: Reinforcement fine-tuning (RFT).

| Metric | Unit | Baseline | Comparison |
|---|---|---|---|
| planning_L2 | m | 0.48 | 0.48 |
| logged_collision | percent | 0.1 | 0.05 |

Aggregation: Source Table 11 reported aggregate, rounded to two decimals.

Boundary: Equal rounded L2 is not proof of equal unrounded error. Policy optimization is not transition-accuracy isolation.

## Analytical example and executable checks

The Section 6.5 merge example stipulates collision probabilities, maneuver penalties and observation errors. Costs are maneuver penalty plus 100 times collision probability. Reference costs are 4.0, 1.2 and 3.0. Model A selects keep speed with regret 2.8; model B selects brake with regret 0. Average observation errors are 0.0149 and 0.0302. The eligible-pair ranking errors are 2/3 and 0, and hazardous false-clearance rates are 1 and 0.

The separately stipulated observation and cost values establish only a logical non-implication. They are not sampled outcomes of the audited systems. data/analytical_example.json stores inputs and expected values.

Run python scripts/validate.py to recompute the example, validate cross-links, calculate numerical changes and run definition/property checks. The script checks strict margin eligibility, half-error ties, unresolved hazard labels, undefined denominators, selective abstention and regret intervals. It does not automatically establish the correctness of source-paper claims.

Confidence intervals are supplied inputs to definition utilities, not estimated by this implementation. Joint coverage, independence and physical validity require separate evidence. Brier and selective-risk tests are deterministic implementation checks, not new driving results.

## Files and correction policy

The companion repository contains the Markdown S1, source-linked JSON register, codebook, numerical transfer file, analytical inputs, query tool, validator, tests and validation report. docs/DATA_DICTIONARY.md explains fields; docs/REPRODUCIBILITY.md describes the verification boundary. Changes should identify a record ID, source version and passage, proposed correction and rationale, and preserve historical releases.